L&Aser for Brazilian soybean

Plant-line laser banding, tractor-scale implements and conditional field capacity.

Two reference scenarios. One 12 m field swath.

Metric	200 ms reference	100 ms scenario
Crop rows	24 × 50 cm	24 × 50 cm
Total modules	72	48
Travel speed	4.29 km/h	8.00 km/h
Effective capacity	3.86 ha/h	7.20 ha/h
Per workable day	61.83 ha	115.20 ha
Fleet for one pass	7	4

Fleet example: 2,000 ha, five workable days, 16 scheduled hours per day and 75% within-day field efficiency. The 100 ms case is a suppression hypothesis at lower weed density, not a matched kill-performance comparison.

These figures are field hectares traversed with laser treatment in crop-row bands. Effective management between the bands is also required. Results do not establish season-long weed control or yield protection.

Rebuilt edition: 9 November 2026. Agronomic and equipment evidence is retained from the study dated 7 November 2026; this is not a new market-availability survey. Electrical accounting is corrected to the all-in allowance clarified in this conversation.

01 | Shared model assumptions

Input	Value / boundary
Working geometry	24 rows × 500 mm = 12 m
Laser bands	152.4 / 177.8 / 203.2 mm
Density inside band	3 / 4 / 5 / 6 weeds/ft² = 32.29 / 43.06 / 53.82 / 64.58 weeds/m²
Total exposure per weed	350 / 250 / 200 / 150 / 100 ms
Serial stages per row	1 / 2 / 3
Retargeting per stage	10 ms
Daily schedule	16 h × 75% = 12 productive-equivalent h
Workable / calendar days	5 / 7
Area in one cohort	2,000 ha
Laser-service utilization	100% ideal mean demand; 80% sensitivity

The full grid contains 3 bands × 4 densities × 5 exposure times × 3 stage counts = 180 cases. The case tables and default workbook grid use a 5 km/h travel cap. The highlighted faster scenario separately uses an 8 km/h cap. [S12]

02 | Crop geometry and planting

Use 45 and 50 cm rows as the main compatibility cases. Embrapa guidance discusses 40–50 cm spacing; the cited Maranhão field example used 50 cm rows and 15 plants per row-metre, equivalent by calculation to 300,000 plants/ha. This example is not a universal cultivar prescription. [S1, S2]

The model adopts 300,000 established plants/ha and a design sensitivity of 250,000–350,000. Average in-row spacing is 7.41 cm at 45 cm row pitch or 6.67 cm at 50 cm. Actual stands include skips and doubles.

Band	Width (mm)	Area share at 45 cm	Area share at 50 cm
6"	152.4	33.87%	30.48%
7"	177.8	39.51%	35.56%
8"	203.2	45.16%	40.64%

Banding reduces the number of weeds assigned to the lasers. It does not change local weed density inside the band.

At 50 cm spacing, an 8-inch band covers 40.64% of the field. At uniform 5 weeds/ft², this is 2.032 laser targets per square foot of field, while local band density remains 5 weeds/ft². Apply the area reduction only once.

03 | Cultivation and no-till

The proposed division of work is mechanical control between crop rows plus precise laser treatment along the plant line. Its suitability depends on soil management, residue, guidance and crop stage. Embrapa identifies operational constraints and widespread direct seeding as reasons mechanical hoeing is less common in soybean. [S3]

A. Combined cultivation and laser bands

Use where the inter-row tools can control weeds without unacceptable crop injury, erosion risk or disruption of the production system. Measure the actual unworked crop strip; laser and mechanical treatment zones must meet or overlap.

B. Residue-preserving laser banding

Keep the laser carrier free of soil-disturbing tools and use an independently effective inter-row program. Banding alone cannot be credited with whole-field control, and ordinary inter-row tillage is not no-till simply because the laser is non-contact.

A Brazilian UNIOESTE study examined camera-guided inter-row hoeing plus manual in-row weeding in organic direct-seeded soybean. Results depended on residue and tools. It supports testing a laser in-row substitute, not a claim that laser performance has already been demonstrated. [S13]

04 | Tractor implement widths

Match the farm’s planter rows and spacing rather than importing vegetable-bed geometry. HORSCH describes a 6–12 m hoe family and a 24-row / 50 cm / 12 m system. LEMKEN describes configurable widths up to 13 m. These are machinery references, not confirmed Brazilian L&Aser combinations. [S8, S9]

Crop rows	Spacing	Field swath	Modules: 1 / 2 / 3 stages
12	45 cm	5.4 m	12 / 24 / 36
18	45 cm	8.1 m	18 / 36 / 54
24	45 cm	10.8 m	24 / 48 / 72
24	50 cm	12.0 m	24 / 48 / 72
30	45 cm	13.5 m	30 / 60 / 90
36	50 cm	18.0 m	36 / 72 / 108

Brazilian manufacturers Agrimec and Moraes list soybean cultivation equipment. Cutting-unit counts must not be mistaken for crop-row counts. Stara and Deere planter specifications support broad working geometries; a planter’s width is not proof of an equally wide cultivator. [S6, S7, S10, S11]

13.5 and 18 m are explicit custom-development extensions. All layouts require wheel-track, transport, row-following and section-control design. Field swath = crop rows × row pitch; transport and frame widths are separate.

05 | Timing and the pass window

Size each pass to the weed cohort and crop stage, not the full growing season. Published interference periods describe crop competition under specific conditions, not permission to take that entire period to complete a laser pass. Scout early and set the deadline locally. [S4, S5]

Planning case	Calendar days	Workable days
Compressed	5	3
Reference	7	5
Extended	10	7

These are educated planning assumptions, not Brazilian agronomic standards. Whole days lost to weather are represented by workable days. Turns, servicing and stops within those days are represented by 75% field efficiency. Do not apply either reduction twice.

Use 3 km/h as a cautious speed case, 5 km/h as the reference and 8 km/h as a conditional faster case. These are design assumptions for this Brazilian study. Actual crop injury, targeting, residue and soil conditions may impose lower limits.

All fleet figures are per pass. Repeat the workload for additional passes and separate differently timed emergence cohorts. A suppression program must include its actual retreatment schedule in any seasonal economic comparison.

06 | Calculation method

Every weed visits every serial stage. The total commanded exposure is divided equally: at 200 ms and three stages, each stage supplies 66.67 ms, then incurs 10 ms retarget time. It is not 200 ms at each stage. [S12]

Stage cycle (s) = total dwell (ms) / stages / 1000 + overhead (ms) / 1000

Targets per row-metre = local density (weeds/m²) × band width (m)

Raw laser speed (km/h) = 3.6 / [targets per row-metre × stage cycle (s)]

Field speed = MIN(travel cap, laser-service utilization × raw laser speed)

Effective ha/h = rows × row spacing (m) × field speed (km/h) × efficiency / 10

Fleet = ROUND UP[area / (ha/h × scheduled h/day × workable days)]

Laser-service utilization and field efficiency represent different limits. The baseline assumes ideal average laser loading (100%); the 80% sensitivity allows scheduling reserve. Neither is a biological-success probability.

The old one-second overspeed adjustment is not used. Residence time, perception, target clustering and crop safety must be assessed separately. Laser density is local to the band and is not discounted again.

07 | Field capacity by width

Both tables use a 203.2 mm band, 16 scheduled h/day, 75% efficiency and five workable days for 2,000 ha. Whole-fleet counts use unrounded capacity.

200 ms · 53.82 weeds/m² · 3 stages · 4.29 km/h

Width	Modules	ha/day	Fleet
5.4 m	36	27.8	15
8.1 m	54	41.7	10
10.8 m	72	55.6	8
12.0 m	72	61.8	7
13.5 m	90	69.6	6
18.0 m	108	92.7	5

100 ms · 32.29 weeds/m² · 2 stages · 8 km/h cap

Width	Modules	ha/day	Fleet
5.4 m	24	51.8	8
8.1 m	36	77.8	6
10.8 m	48	103.7	4
12.0 m	48	115.2	4
13.5 m	60	129.6	4
18.0 m	72	172.8	3

The wider 13.5 and 18 m cases are custom-development scenarios, not existing integrated product specifications.

08 | Operating sensitivity

For the 12 m, 200 ms reference, changes to available hours or usable capacity can change the fleet materially. Each row below changes only one assumption.

Changed assumption	ha/workable day	Fleet
Reference	61.83	7
12 hours/day	46.37	9
80% laser utilization	49.46	9
60% field efficiency	49.46	9
3 workable days	61.83	11

When a third stage helps

At 100 ms, 3 weeds/ft² and an 8-inch band, two stages have a 9.14 km/h laser ceiling. Three stages raise it to 12.66 km/h. At an 8 km/h travel cap, both give 115.2 ha/day in the ideal case.

At 80% laser-service utilization, two stages are limited to 7.32 km/h. A third can restore the assumed 8 km/h target. It supplies operating reserve rather than automatically increasing useful field speed.

The 6-inch / 100 ms / 3 weeds per ft² / 3-stage case calculates to a 16.88 km/h laser ceiling. This is not a validated cultivator speed.

09 | Power and mass boundaries

Electrical power is an all-in laser-system allowance. Mass is a module-only allowance. These boundaries are different.

Use 1.1 kW per single-channel module, with shared laser-system loads allocated: laser, cooling, lighting, HMI, compute, electronics and safety functions. Do not add these included electrical loads a second time. This corrects the earlier report's wording. [S12]

Configuration	72 modules	48 modules
Laser-system input allowance	79.2 kW	52.8 kW
Module mass only	1,800 kg	1,200 kg
Reference field capacity	3.86 ha/h	7.20 ha/h
Budgeted electrical energy	20.50 kWh/ha	7.33 kWh/ha

Energy per hectare above holds the full electrical allowance throughout each scheduled hour and divides by effective field capacity. It is not metered energy. Propulsion, cultivation draft, generator losses and genuinely additional loads remain separate.

Complete implement mass must also include the frame, cultivator, generation equipment, shielding, cooling hardware and other integration parts. The 25 kg/module assumption does not establish delivered system weight.

This rebuilt package does not turn the separately discussed Carbon generator extrapolation into a verified specification or add it to the soybean throughput model.

10 | Validation before deployment

Proposed biological outcome	Total commanded exposure
Immediate kill	350 ms
Kill within 24 h	250 ms
Kill within 48 h	200 ms
Stunt	150 ms
Suppress	100 ms

These labels are treatment hypotheses, not validated Brazilian soybean-field results. Verify weed species and size, delivered dose, regrowth, reproduction, crop injury and the equivalence of split versus continuous exposure.

Validate perception latency, target identity across stages, accessible treatment-zone length, loaded-row hotspots and safe exclusions around crop tissue. Spare capacity in one row may not be transferable to another.

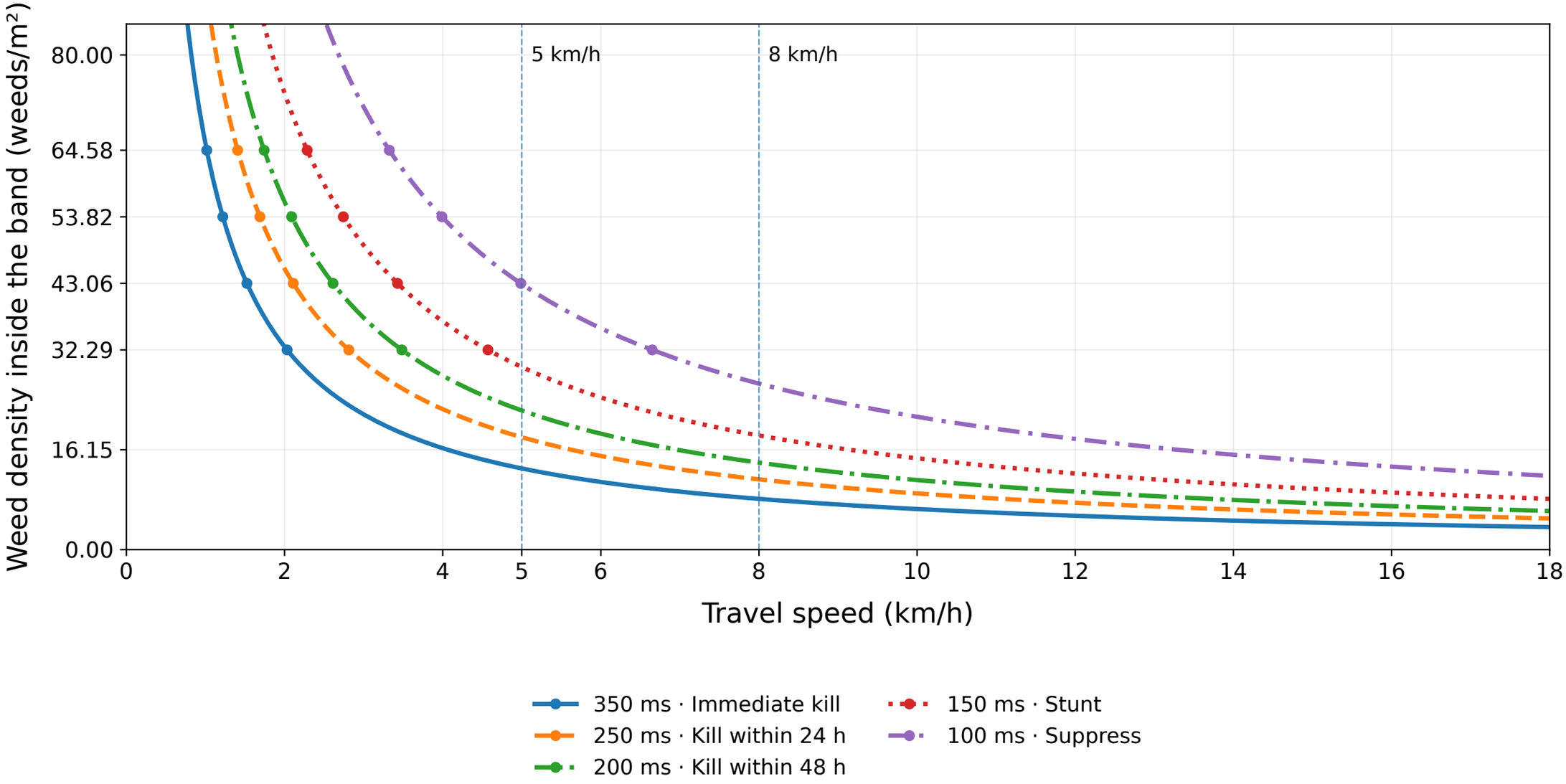
Test terrain following, section control, shielding, dust, residue ignition risk, fire response, cooling and tractor compatibility. Record successful treatments and inter-row escapes separately from hectares traversed.

A high-capacity suppression pass is not proof of season-long control. No current sales offer, manufacturer partnership, purchase price, herbicide saving, payback or yield benefit is asserted.

L&Aser | Brazilian soybean

Total treatment band: 152.4 mm (6")

1 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

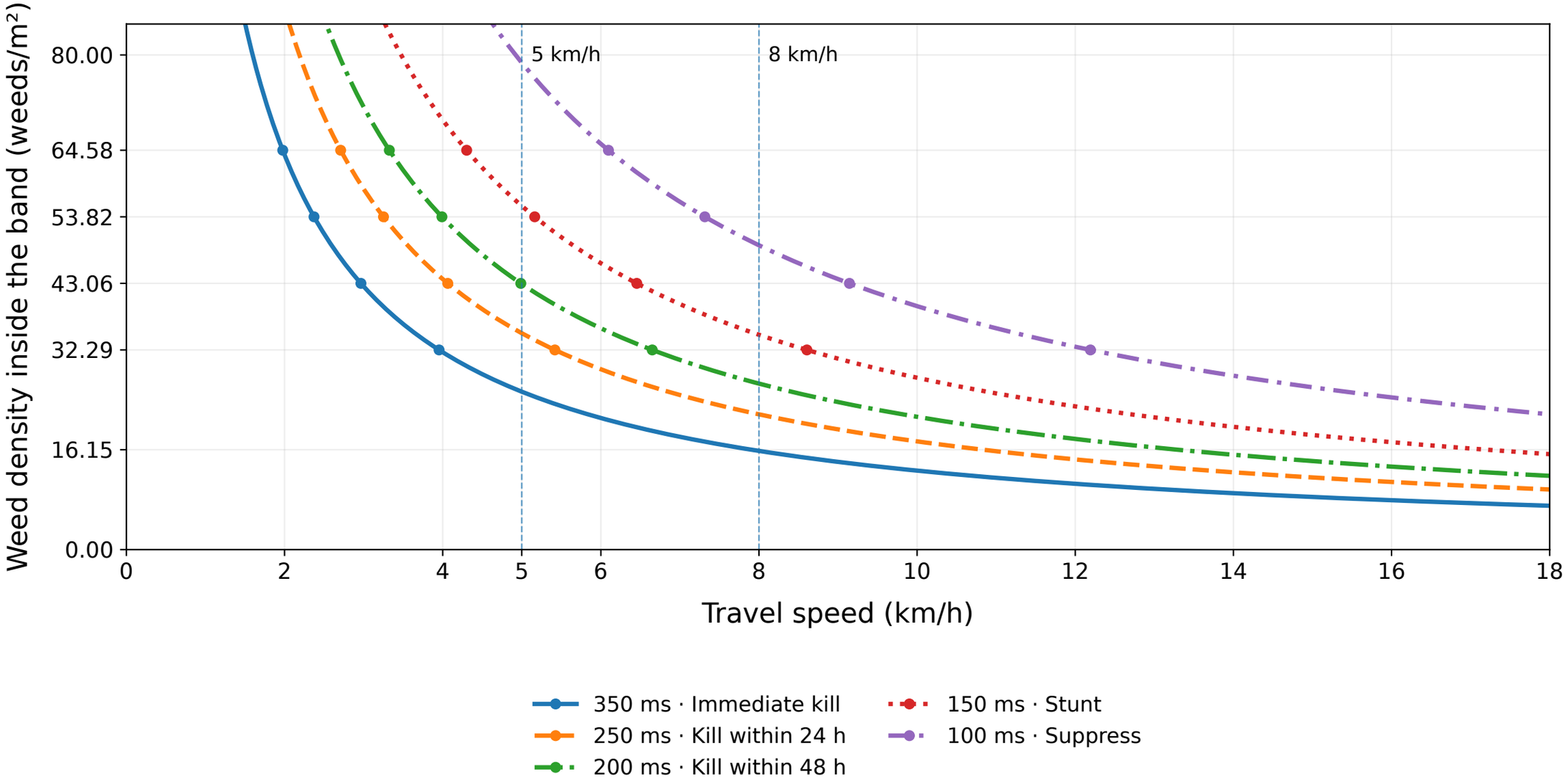
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 152.4 mm (6")

2 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

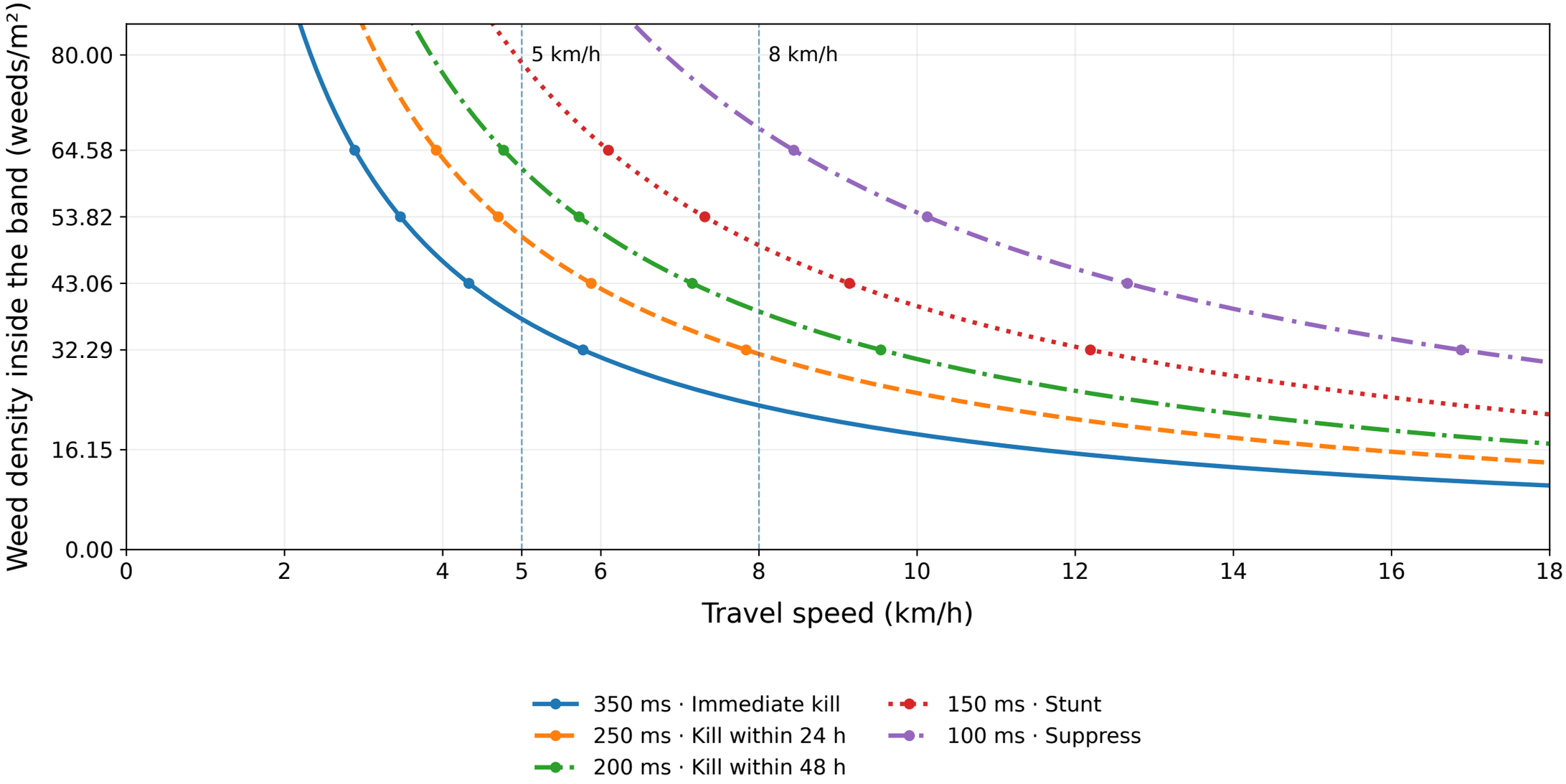
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 152.4 mm (6")

3 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

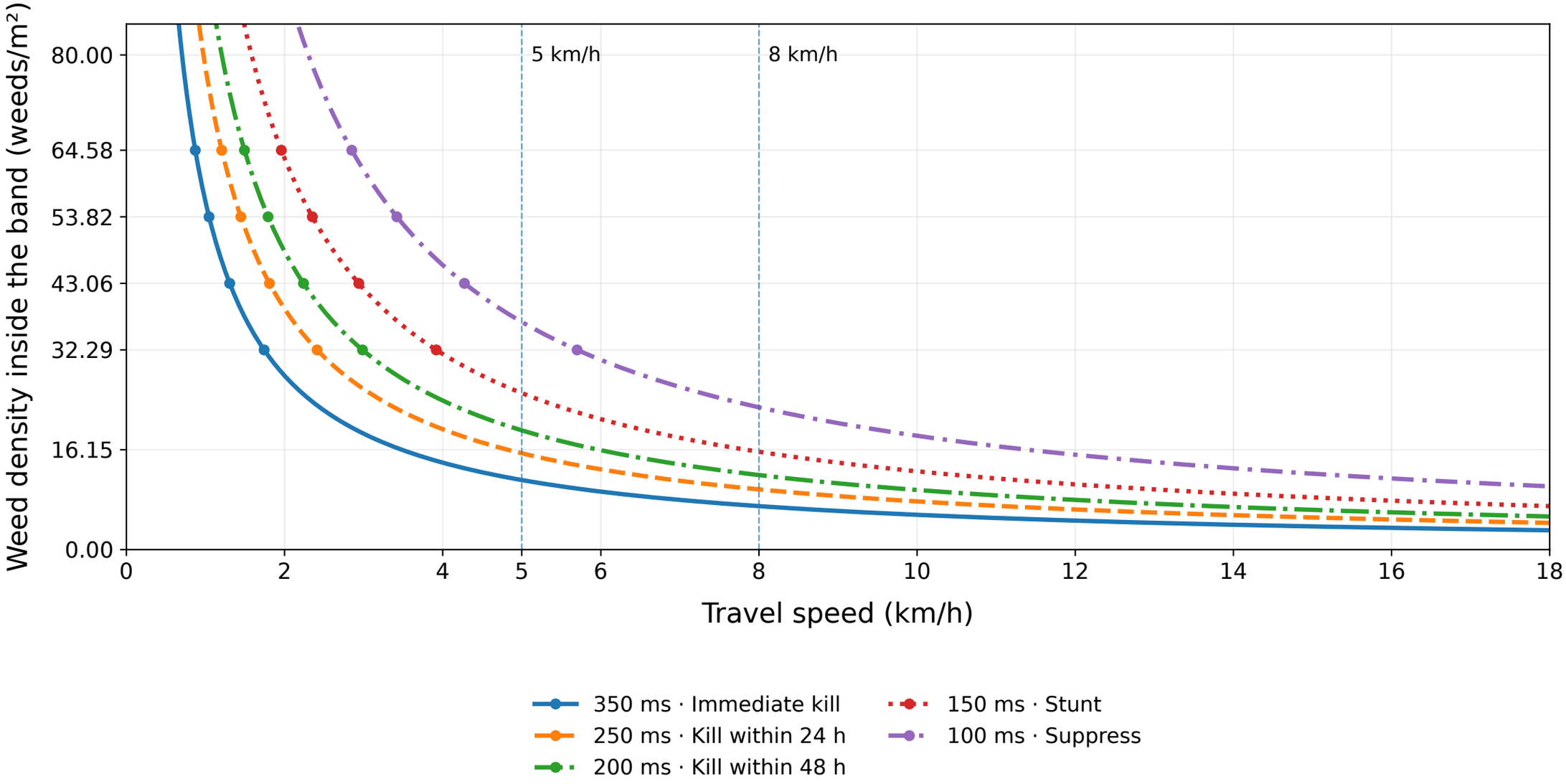
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 177.8 mm (7")

1 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

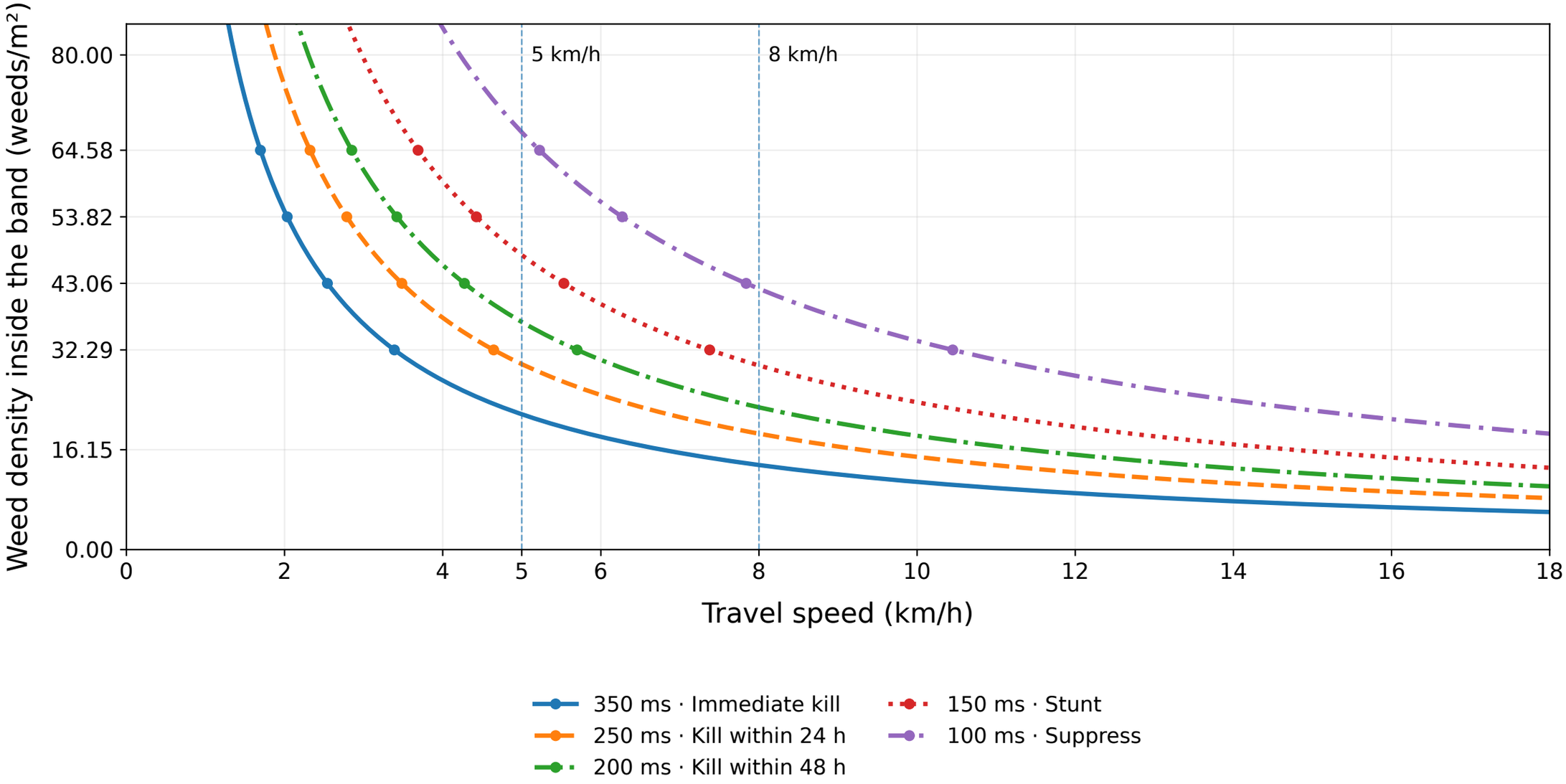
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 177.8 mm (7")

2 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

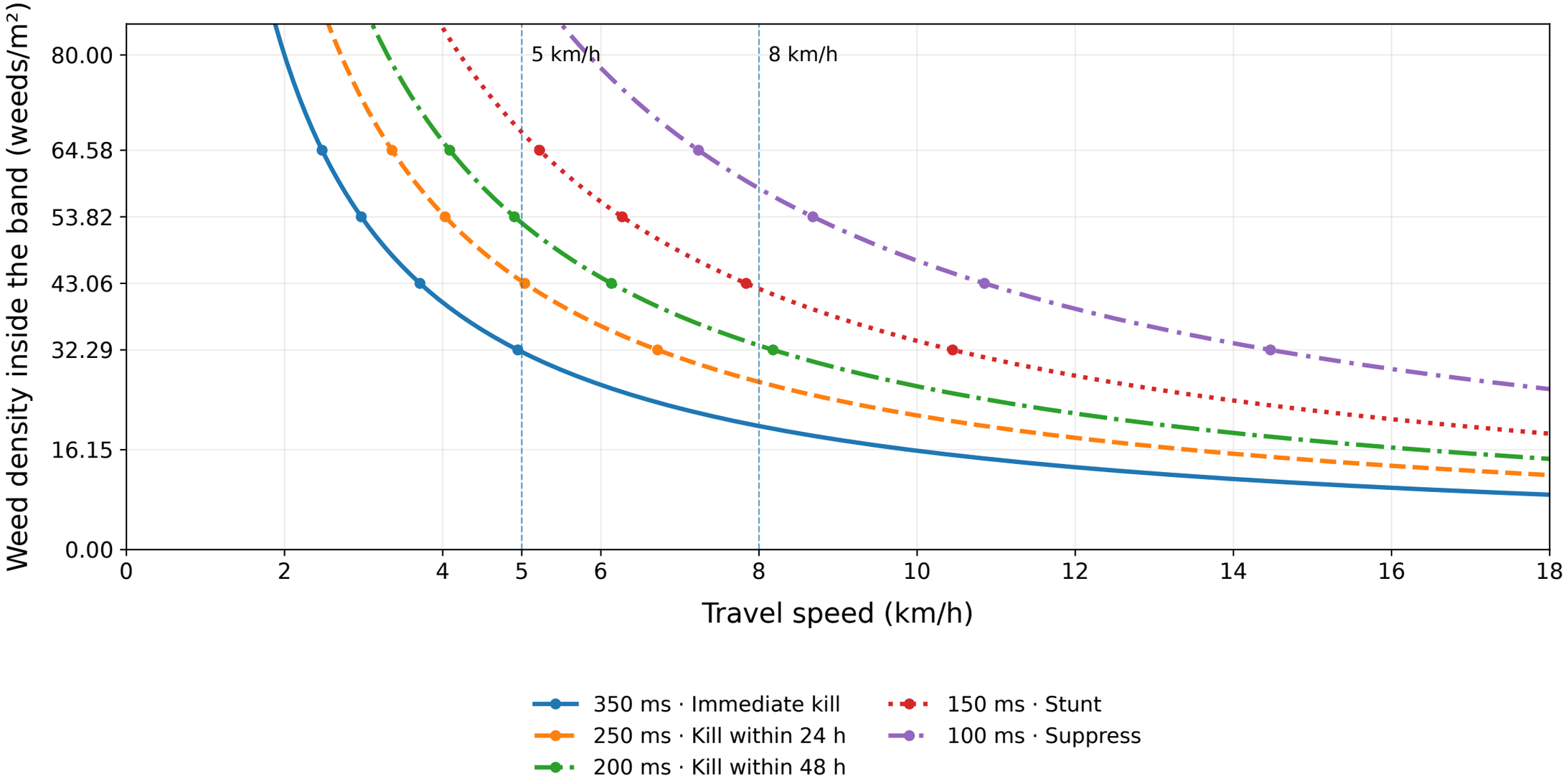
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 177.8 mm (7")

3 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

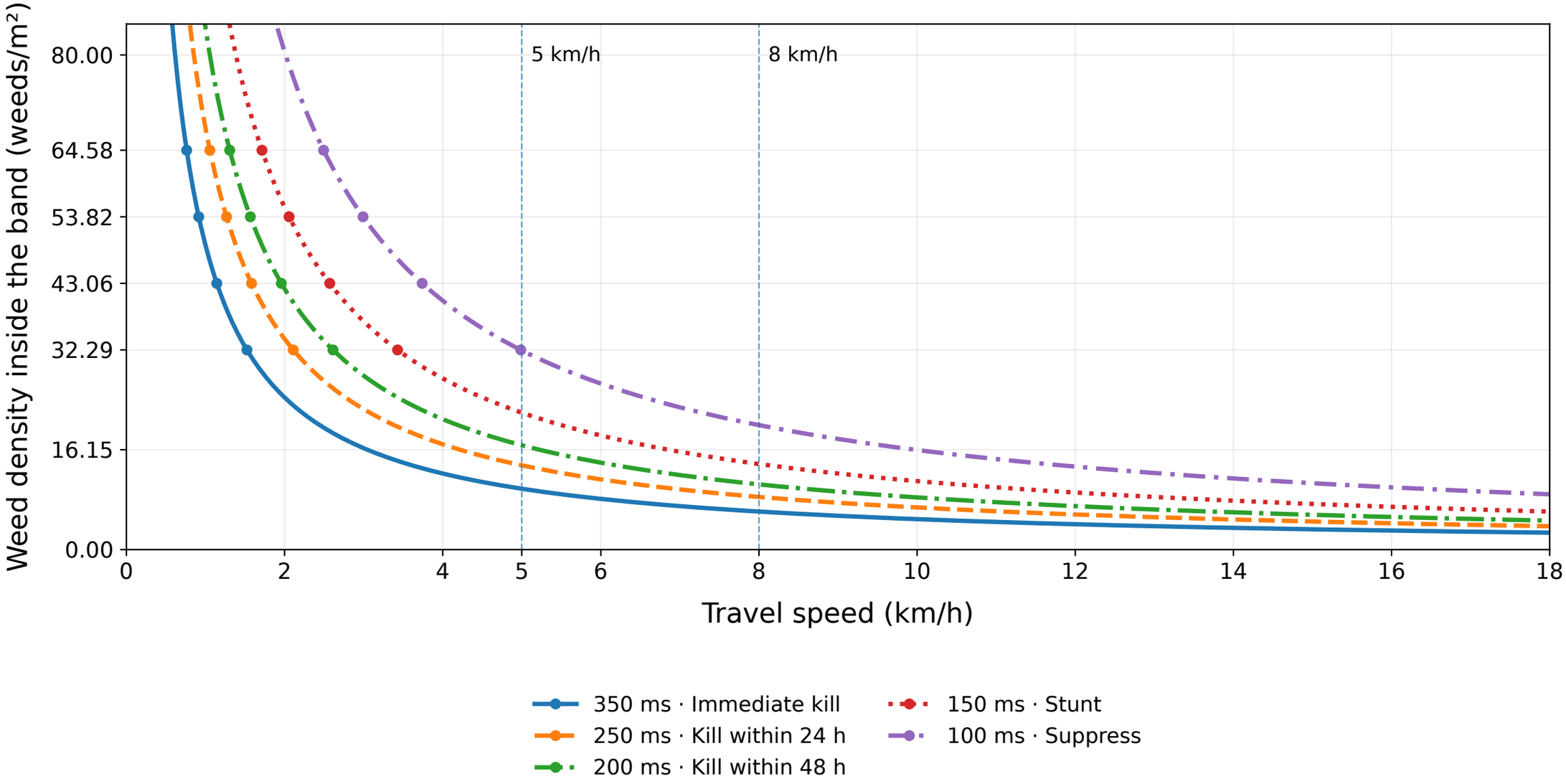
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 203.2 mm (8")

1 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

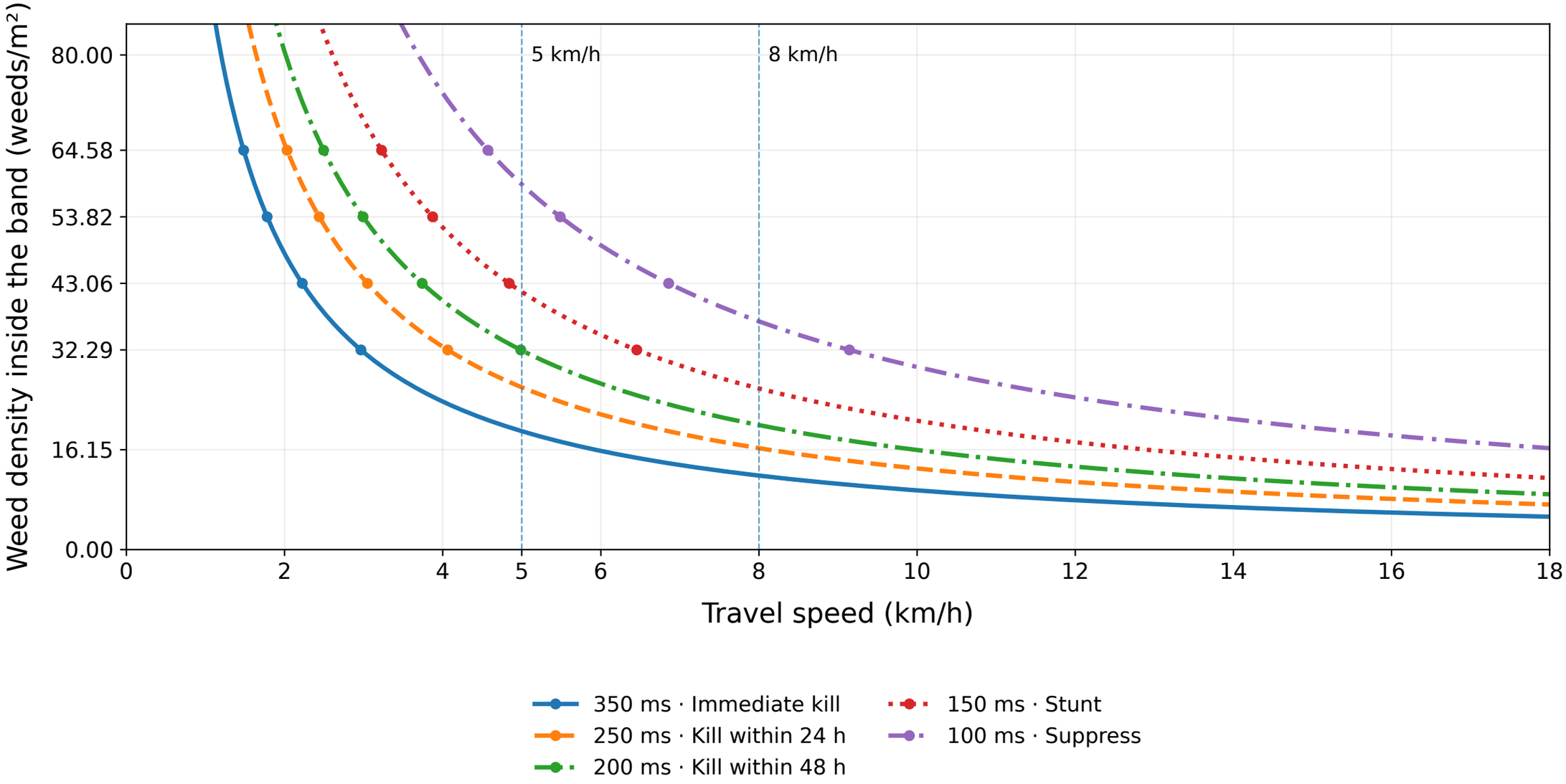
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 203.2 mm (8")

2 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

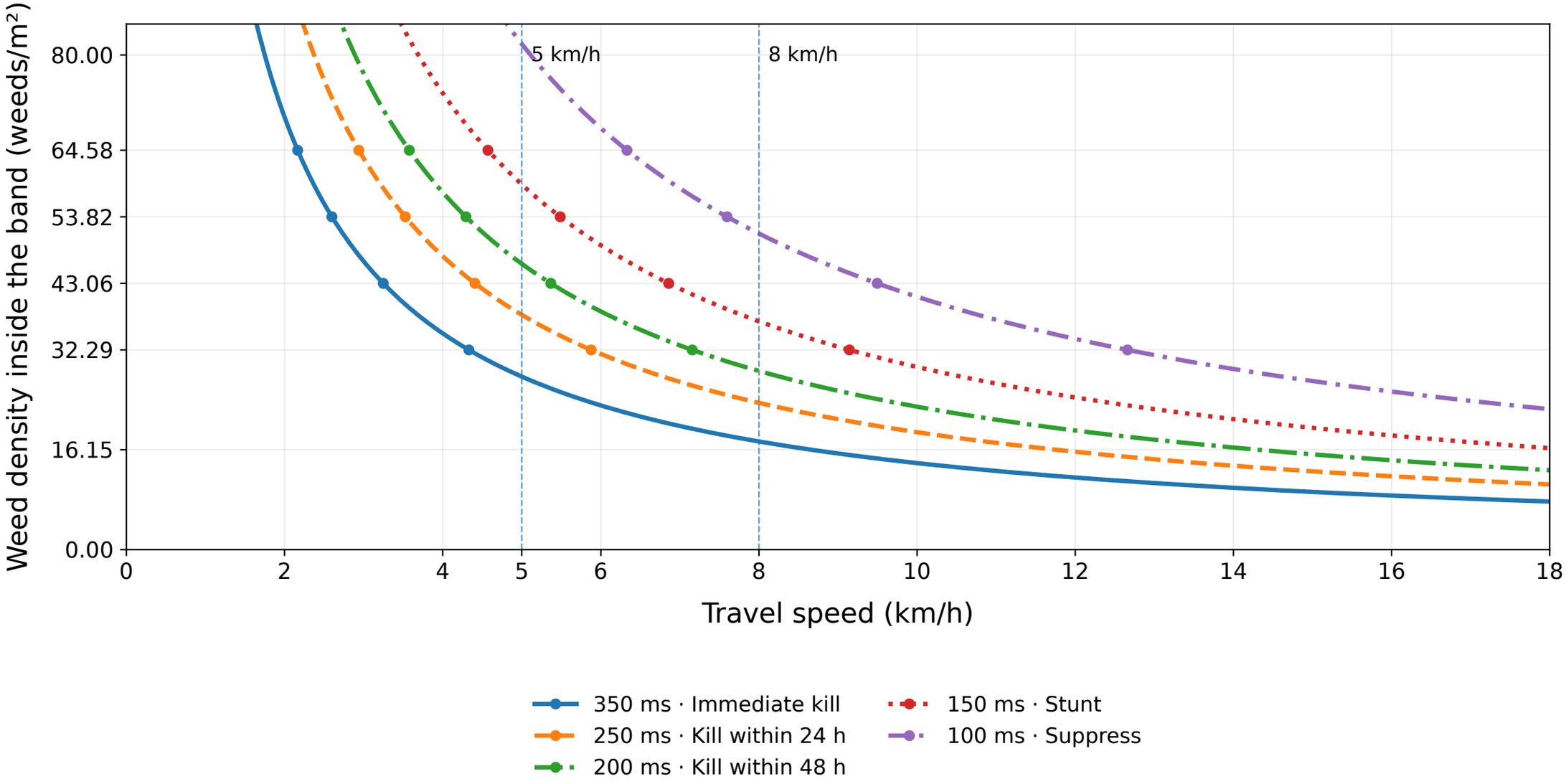
Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

L&Aser | Brazilian soybean

Total treatment band: 203.2 mm (8")

3 serial modules per crop row



Biological outcomes are hypotheses. Curves show calculated laser-service ceilings.

Total exposure is split between stages; each stage adds 10 ms retarget time.

Dashed lines: 5 and 8 km/h planning caps, not validated operating speeds.

Numerical cases | 152.4 mm

1 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	2.03	2.03	29.26	14
350	43.06	1.52	1.52	21.95	19
350	53.82	1.22	1.22	17.56	23
350	64.58	1.02	1.02	14.63	28
250	32.29	2.81	2.81	40.51	10
250	43.06	2.11	2.11	30.39	14
250	53.82	1.69	1.69	24.31	17
250	64.58	1.41	1.41	20.26	20
200	32.29	3.48	3.48	50.16	8
200	43.06	2.61	2.61	37.62	11
200	53.82	2.09	2.09	30.10	14
200	64.58	1.74	1.74	25.08	16
150	32.29	4.57	4.57	65.84	7
150	43.06	3.43	3.43	49.38	9
150	53.82	2.74	2.74	39.50	11
150	64.58	2.29	2.29	32.92	13
100	32.29	6.65	5.00	72.00	6
100	43.06	4.99	4.99	71.82	6
100	53.82	3.99	3.99	57.46	7
100	64.58	3.33	3.33	47.88	9

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 152.4 mm

2 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	3.95	3.95	56.94	8
350	43.06	2.97	2.97	42.70	10
350	53.82	2.37	2.37	34.16	12
350	64.58	1.98	1.98	28.47	15
250	32.29	5.42	5.00	72.00	6
250	43.06	4.06	4.06	58.52	7
250	53.82	3.25	3.25	46.82	9
250	64.58	2.71	2.71	39.01	11
200	32.29	6.65	5.00	72.00	6
200	43.06	4.99	4.99	71.82	6
200	53.82	3.99	3.99	57.46	7
200	64.58	3.33	3.33	47.88	9
150	32.29	8.61	5.00	72.00	6
150	43.06	6.45	5.00	72.00	6
150	53.82	5.16	5.00	72.00	6
150	64.58	4.30	4.30	61.96	7
100	32.29	12.19	5.00	72.00	6
100	43.06	9.14	5.00	72.00	6
100	53.82	7.32	5.00	72.00	6
100	64.58	6.10	5.00	72.00	6

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 152.4 mm

3 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	5.78	5.00	72.00	6
350	43.06	4.33	4.33	62.37	7
350	53.82	3.47	3.47	49.90	9
350	64.58	2.89	2.89	41.58	10
250	32.29	7.84	5.00	72.00	6
250	43.06	5.88	5.00	72.00	6
250	53.82	4.70	4.70	67.72	6
250	64.58	3.92	3.92	56.43	8
200	32.29	9.54	5.00	72.00	6
200	43.06	7.16	5.00	72.00	6
200	53.82	5.72	5.00	72.00	6
200	64.58	4.77	4.77	68.70	6
150	32.29	12.19	5.00	72.00	6
150	43.06	9.14	5.00	72.00	6
150	53.82	7.32	5.00	72.00	6
150	64.58	6.10	5.00	72.00	6
100	32.29	16.88	5.00	72.00	6
100	43.06	12.66	5.00	72.00	6
100	53.82	10.13	5.00	72.00	6
100	64.58	8.44	5.00	72.00	6

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 177.8 mm

1 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	1.74	1.74	25.08	16
350	43.06	1.31	1.31	18.81	22
350	53.82	1.05	1.05	15.05	27
350	64.58	0.87	0.87	12.54	32
250	32.29	2.41	2.41	34.73	12
250	43.06	1.81	1.81	26.05	16
250	53.82	1.45	1.45	20.84	20
250	64.58	1.21	1.21	17.36	24
200	32.29	2.99	2.99	43.00	10
200	43.06	2.24	2.24	32.25	13
200	53.82	1.79	1.79	25.80	16
200	64.58	1.49	1.49	21.50	19
150	32.29	3.92	3.92	56.43	8
150	43.06	2.94	2.94	42.32	10
150	53.82	2.35	2.35	33.86	12
150	64.58	1.96	1.96	28.22	15
100	32.29	5.70	5.00	72.00	6
100	43.06	4.28	4.28	61.56	7
100	53.82	3.42	3.42	49.25	9
100	64.58	2.85	2.85	41.04	10

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 177.8 mm

2 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	3.39	3.39	48.81	9
350	43.06	2.54	2.54	36.60	11
350	53.82	2.03	2.03	29.28	14
350	64.58	1.69	1.69	24.40	17
250	32.29	4.64	4.64	66.88	6
250	43.06	3.48	3.48	50.16	8
250	53.82	2.79	2.79	40.13	10
250	64.58	2.32	2.32	33.44	12
200	32.29	5.70	5.00	72.00	6
200	43.06	4.28	4.28	61.56	7
200	53.82	3.42	3.42	49.25	9
200	64.58	2.85	2.85	41.04	10
150	32.29	7.38	5.00	72.00	6
150	43.06	5.53	5.00	72.00	6
150	53.82	4.43	4.43	63.73	7
150	64.58	3.69	3.69	53.11	8
100	32.29	10.45	5.00	72.00	6
100	43.06	7.84	5.00	72.00	6
100	53.82	6.27	5.00	72.00	6
100	64.58	5.23	5.00	72.00	6

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 177.8 mm

3 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	4.95	4.95	71.28	6
350	43.06	3.71	3.71	53.46	8
350	53.82	2.97	2.97	42.77	10
350	64.58	2.48	2.48	35.64	12
250	32.29	6.72	5.00	72.00	6
250	43.06	5.04	5.00	72.00	6
250	53.82	4.03	4.03	58.04	7
250	64.58	3.36	3.36	48.37	9
200	32.29	8.18	5.00	72.00	6
200	43.06	6.13	5.00	72.00	6
200	53.82	4.91	4.91	70.66	6
200	64.58	4.09	4.09	58.89	7
150	32.29	10.45	5.00	72.00	6
150	43.06	7.84	5.00	72.00	6
150	53.82	6.27	5.00	72.00	6
150	64.58	5.23	5.00	72.00	6
100	32.29	14.47	5.00	72.00	6
100	43.06	10.85	5.00	72.00	6
100	53.82	8.68	5.00	72.00	6
100	64.58	7.23	5.00	72.00	6

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 203.2 mm

1 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	1.52	1.52	21.95	19
350	43.06	1.14	1.14	16.46	25
350	53.82	0.91	0.91	13.17	31
350	64.58	0.76	0.76	10.97	37
250	32.29	2.11	2.11	30.39	14
250	43.06	1.58	1.58	22.79	18
250	53.82	1.27	1.27	18.23	22
250	64.58	1.06	1.06	15.19	27
200	32.29	2.61	2.61	37.62	11
200	43.06	1.96	1.96	28.22	15
200	53.82	1.57	1.57	22.57	18
200	64.58	1.31	1.31	18.81	22
150	32.29	3.43	3.43	49.38	9
150	43.06	2.57	2.57	37.03	11
150	53.82	2.06	2.06	29.63	14
150	64.58	1.71	1.71	24.69	17
100	32.29	4.99	4.99	71.82	6
100	43.06	3.74	3.74	53.87	8
100	53.82	2.99	2.99	43.09	10
100	64.58	2.49	2.49	35.91	12

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 203.2 mm

2 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	2.97	2.97	42.70	10
350	43.06	2.22	2.22	32.03	13
350	53.82	1.78	1.78	25.62	16
350	64.58	1.48	1.48	21.35	19
250	32.29	4.06	4.06	58.52	7
250	43.06	3.05	3.05	43.89	10
250	53.82	2.44	2.44	35.11	12
250	64.58	2.03	2.03	29.26	14
200	32.29	4.99	4.99	71.82	6
200	43.06	3.74	3.74	53.87	8
200	53.82	2.99	2.99	43.09	10
200	64.58	2.49	2.49	35.91	12
150	32.29	6.45	5.00	72.00	6
150	43.06	4.84	4.84	69.71	6
150	53.82	3.87	3.87	55.77	8
150	64.58	3.23	3.23	46.47	9
100	32.29	9.14	5.00	72.00	6
100	43.06	6.86	5.00	72.00	6
100	53.82	5.49	5.00	72.00	6
100	64.58	4.57	4.57	65.84	7

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Numerical cases | 203.2 mm

3 serial stage(s) per crop row. Default: 12 m swath; 5 km/h travel cap; 75% efficiency; 16 h/day; five workable days; 2,000 ha.

Total ms	Weeds/m²	Laser km/h	Field km/h	ha/day	Fleet
350	32.29	4.33	4.33	62.37	7
350	43.06	3.25	3.25	46.78	9
350	53.82	2.60	2.60	37.42	11
350	64.58	2.17	2.17	31.19	13
250	32.29	5.88	5.00	72.00	6
250	43.06	4.41	4.41	63.49	7
250	53.82	3.53	3.53	50.79	8
250	64.58	2.94	2.94	42.32	10
200	32.29	7.16	5.00	72.00	6
200	43.06	5.37	5.00	72.00	6
200	53.82	4.29	4.29	61.83	7
200	64.58	3.58	3.58	51.52	8
150	32.29	9.14	5.00	72.00	6
150	43.06	6.86	5.00	72.00	6
150	53.82	5.49	5.00	72.00	6
150	64.58	4.57	4.57	65.84	7
100	32.29	12.66	5.00	72.00	6
100	43.06	9.50	5.00	72.00	6
100	53.82	7.60	5.00	72.00	6
100	64.58	6.33	5.00	72.00	6

Density is local to the band. Fleet counts use unrounded calculations. Exposure labels are hypotheses.

Sources and traceability

Retained research register from 7 November 2026. Source titles are clickable. Full URLs are also supplied in the workbook and browser edition.

[S1] Embrapa Ageitec — Aspectos a serem considerados

Embrapa: 40–50 cm row-spacing guidance; local cultivar conditions govern.

[S2] Chaves et al. / Embrapa Meio-Norte (2026)

Maranhão field example: 0.5 m rows, 15 plants/m; not a universal prescription.

[S3] Embrapa Ageitec — Manejo de plantas daninhas

Operational limitations of mechanical hoeing and direct-seeding context.

[S4] Embrapa Ageitec — Períodos de convivência

Crop competition timing is not the number of days allowed to finish a machinery pass.

[S5] Albrecht et al. — X Congresso Brasileiro de Soja (2025), abstract 250

Site-specific goosegrass interference study; printed p. 280.

[S6] Agrimec — ROTACARP

Brazilian cultivator; distinguish cutting units from planted crop rows.

[S7] Moraes — VIBRO

Brazilian inter-row equipment example; not a verified integrated laser system.

[S8] HORSCH — Transformer VF

6–12 m hoe family and a 24-row / 50 cm / 12 m user configuration.

[S9] LEMKEN — EC-Weeder

Configurable global hoeing platform, widths up to 13 m.

[S10] John Deere Brasil — 2130

Planter geometry: 30 rows at 45 cm or 26 at 50 cm; not cultivator availability.

[S11] Stara Brasil — Absoluta

Planter configurations include 24 or 36 rows at 50 cm; large laser implementations remain proposed.

[S12] L&Aser sunflower model and founder inputs

User-provided: 180 cases, split total exposure, 10 ms/stage, 75% efficiency, 16 h/day. Electrical allocation clarified in this conversation.

[S13] UNIOESTE — Canavessi, camera-guided soybean hoeing

Organic direct-seeded soybean: mechanical inter-row control and manual in-row weeding. No laser trial.